1 · Scientific question and representation contract



Same object identity is the bridge between different feature spaces.

Question — Do frozen models organize the **same galaxies** similarly across surveys, and does that geometry support the **same unchanged physical predictor**? Optical/infrared/spectral observations measure different aspects of one object. Similar neighborhoods are an observational relationship; physical transfer is a separate test.

Shapes — Images B×C×H×W → patches/tokens B×T×D → pooled B×D. Sequence adapters accept B×C×T; tabular B×F. Output: embed_for_mode→B×D; layerwise→ordered {stage:B×D}. Coordinates are learned features, not automatically mass, redshift or size.

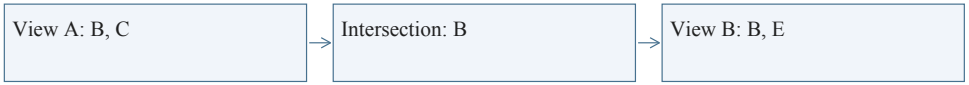
Hook code — ModelAdapter._capture_module_outputs installs selected forward hooks, accepts first tuple tensor, pools, float-casts/detaches, runs under no_grad and removes hooks in finally. Generic pooling averages spatial axes for 4D and tokens for 3D. Prevents leaked hooks; rank alone cannot identify batch axis or exclude padding/special tokens.

Modality tradeoff — fit_length right-pads with zeros+mask or keeps first T samples. Chronos takes channel 0 and mean-pools encoder tokens: no multichannel fusion is implied. Real light-curve/catalog wiring remains separate from an adapter smoke test.

2 · Cosine, neighbors and local agreement

$$\cos(x,y)=x^{\mathsf{T}}y/[\mathrm{L2norm}(x)\ \mathrm{L2norm}(y)]$$
$$Z_norm[i]=Z[i]/[\mathrm{L2norm}(Z[i])+\varepsilon];\ \mathrm{MKNN}=(1/N)\ \Sigma_i\ |N_A(i)\cap N_B(i)|/k$$

Concrete meaning — [1,2,3] and [2,4,6] have cosine 1 because proportions match. For one galaxy, neighbor lists [B,C] and [B,E] share one of two IDs: overlap=.5. Compare within-view neighbors, then identities across views; d_A may differ from d_B. "Mutual" here means cross-space overlap, not reciprocal edges.



For galaxy A: |intersection|/k = 1/2. Compare object IDs, not coordinates.

Chance and preprocessing — Uniform independent neighbor lists excluding self have expected overlap fraction k/(N−1). Optional global-percentile clipping suppresses activation outliers but changes geometry. CPU mknn signature defaults k=10, truncate_percentile=100 (disabled), despite a stale docstring mentioning 95. Paper uses k=20; record the actual setting.

Failure controls — Zero vectors have no meaningful direction; epsilon merely avoids division by zero. Reject/mark zero and nonfinite rows with a common paired mask. Explicitly exclude self, define ties and compare CPU/GPU neighbors on a hand-checkable fixture. Self-MKNN=1 and shuffled-pair controls test different failures.

3 · Depth: state versus update

Stage semantics — State h_l is accumulated representation; update Δh_l=h_l−h_(l−1) measures change. Subtract only compatible widths/coordinates/pooling. Blocks hook coarse modules; residual hooks non-leaves; leaves hooks terminal operations. Named-module traversal order is not proof of a sequential residual stream. Save stage names, execution order and shapes; more inspected stages increase selection opportunity.

4 · Pairing, splits and geometric alternatives

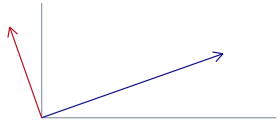
Identity before shape — Matched surveys must share unique galaxy IDs and exact row order. Extractor accumulates paired modes and drops columns whose row count differs from the majority (catches token-vs-batch errors). Equal row counts do not prove correspondence. The shown Parquet writer does not explicitly add object IDs/manifests: persist IDs, filter masks and revisions as a strengthening.

$$\text{Linear CKA} = \|X_c^{\mathsf{T}}Y_c\|^2_F\ /\ (\|X_c^{\mathsf{T}}X_c\|_F\ \|Y_c^{\mathsf{T}}Y_c\|_F)$$
$$X_c = X - \text{column means};\ Y_c \text{ likewise}$$

Metric distinctions — MKNN: local identity-neighborhood agreement. CKA: centered relational similarity, invariant to orthogonal transforms/isotropic scale. Procrustes: fits an orthogonal alignment. Exact-object retrieval: does the true paired object rank highly? RSA correlates pairwise distances; Jaccard divides neighbor intersection by union. These answer related but nonidentical questions.

Implementation cost — kernel.cka builds N×N centered Gram matrices (linear/RBF options). GPU worker uses feature-space linear CKA; denominator reused over permutations. Intermediate sizes include d_A^2, d_B^2 and d_Ad_B: favorable when d much less than N, expensive for wide output heads. Default worker excludes lm_head and d>16,384; exclusion is unavailable, not zero.

5 · Prediction, transfer and learned adaptation



x=[2,1] → R(x)=[−1,2]
Fixed predictor "read coordinate 1": 2 → −1.
Rotate **every** object: angles/neighbors survive; the old readout need not.

$$\text{Ridge: minimize } \|Xw+b-y\|^2 + \lambda\|w\|^2 \text{ (Euclidean norms)}$$
$$R^2 = 1 - \Sigma(y-\hat{y})^2\ /\ \Sigma(y-\bar{y}_{test})^2$$

Readout test — Fit preprocessing and predictor on allowed source rows; choose settings/stage on validation roles; apply unchanged to the target test. R^2=−1.5 means squared error is 2.5× the test-mean baseline, not negative accuracy. A rotation preserves angles/neighbors but changes a fixed coordinate readout.

Code vs paper — physics.linear_probe L2-normalizes, optionally fits PCA inside each fold, then uses OLS; returns mean/std/folds despite a float annotation. That is within-view decodability, not automatically the paper ridge transfer protocol. Its PCA cap uses full N rather than fold N, so requested components can exceed training-fold rank.

Adaptation controls — PCA retains high variance; CCA finds cross-view correlated directions; Procrustes learns a rotation/reflection. Fit on training pairs only. Target refitting or learned alignment is adaptation, a different claim from unchanged transfer. High-dimensional CCA can overfit; tune dimension/ridge on validation, test once, and compare shuffled-pair/random-feature/within-view baselines.

Leakage rule — Galaxy-level grouping, shared row masks, frozen preprocessing and disjoint fit/selection/test roles matter more than a seed. Never fit PCA/normalization statistics on the full test-containing matrix when the protocol requires training-only fitting. Report sample counts after filtering.

6 · Permutations and the searched maximum

$$p = (1 + \#\{\text{null} \geq \text{observed}\}) / (B + 1)$$
Worker calibrated score = clip[(observed−τ)/(1−τ), 0, 1]

Graph reuse — Build both neighbor graphs once. For permutation π, relabel graph B as π^(-1)(N_B(π(i))). Both query rows and neighbor IDs must change; shuffling rows alone leaves mismatched labels. This keeps within-view geometry while destroying cross-view correspondence and avoids repeated pairwise search.

```
perm_inv[perm] = arange(N)
B_shuffled = perm_inv[B_neighbors[perm]]
```

Selection correction — The worker calibrates each layer pair separately using pair-specific seeds. Taking its best cell is **not** a maximum-over-depth permutation test. To test a searched peak, apply the same search to each shuffled curve with a common permutation across depth, then compare maxima. Paper: k=20, B=9,999 depth-max permutations; worker defaults: k=10, B=1,000 (minimum p≈.001).

7 · Extraction and artifact engineering

extract_all_layers — Uses no_grad, collects CPU tensors per stage/mode, concatenates and writes wide Parquet. Precomputed DESI/SDSS embeddings bypass model forward. Saved embeddings let metrics/probes change without repeating inference. But this accumulates all batches before writing; it is not a bounded streaming writer.

Stored float32 activation bytes ≈ 4N Σ_(stages)d_l
Include both views, concatenation copies and metadata in RAM estimates.

Worker cache — Download one Parquet to local scratch; read selected columns. OrderedDict LRU defaults to 64 decimal GB and keeps at least two columns, so that budget can be exceeded by two large arrays. Persist model/checkpoint/data/code revisions, IDs, pooling, stage map, dtype, seeds and metric settings. Streaming row groups/shards can bound host RAM.

8 · GPU memory and exact-search complexity



Float32 scores only. One retained N×20 int64 graph: 16 MB.

_build_knn_chunked — Compute C×N cosine scores; mask local row/global self column to −∞; topk IDs into N×k int64 table; discard scores. _mknn_overlap_sparse broadcasts to N×k×k booleans, then counts matches. Storage is sparse; equality work still grows with k².

Worked budget — N=100,000, C=4096, k=20: full float32 scores 40 GB; chunk 1.6384 GB; one int64 graph 16 MB; two graphs 32 MB; N×k² Boolean intermediate ~40 MB. Embeddings/top-k workspace/CKA add more. Exact neighbor search still O(N²d); chunking saves peak storage, not quadratic arithmetic.

Alternatives — Smaller chunks reduce GPU peak but increase launches. Exact FAISS flat search is still exact; approximate indices need recall@k and MKNN-bias checks against exact subsets. TF32 can alter close-score rankings: compare strict float32 stability. For large k, sorted-set intersections can avoid k² matching.

9 · Workers, failure recovery and resource claims

Task lifecycle — Discover survey/model → claim → download → load columns → compare/validate → write temp → rename → upload done → release claim. SLURM example requests 1 GPU, 8 CPUs, 512 GB host RAM, 24h. Independent pair partitions flatten i×n_B+j; partial files need gap/duplicate checks before merge. This is task parallelism, not model sharding.

Guarantees — Local O_CREAT|O_EXCL claims atomically on a shared filesystem; stale reclaim can race with a live old owner. Remote running-marker upload is not compare-and-swap. PID-specific temp then rename prevents partial local results, but does not make remote completion exactly once. Add lease fencing/ownership, unique attempt IDs, checksums and idempotent publication.

10 · Reported findings and how to read the figures

Paper-linked result	Interpretation
36 models / 11 families; 139/139 comparisons	Significant local agreement somewhere in depth
46 stronger update peaks; 71% earlier update peak	Updates differ from accumulated states; not causal proof
8/8 geometry-selected image transfer tests worse	Geometry selection insufficient for best held-out R²
31/31 terminal redshift transfers R²<0	Unchanged readout fails test-mean baseline
Within-view mean R² .503	Useful decodable signal within a survey
Cross-view median −1.43; mean −2.98	Coordinate/readout transfer is a harder requirement

Figure reading — Fig.1: state/update curves over normalized depth; final layer need not peak. Fig.2: geometry–prediction association depends on target; best-vs-best correlation is not a selection policy. Fig.3: negative geometry-selection regret and negative cross-survey readout despite positive within-view prediction. Counts are retained from paper-linked notes, not recomputed here.

11 · Contribution, interpretability and reproduction

Documented contribution — ARAVIND.md attributes time-series/spectrum/tabular adapters and CPU-friendly device/determinism support. Real-weight CPU checks reported for Chronos, PatchTST/PatchTSMixer, Moirai and Spender. MOMENT dependency conflict; TabPFN gated weights; 39 stub tests reported. Dataset wiring remained. Stub interface test ≠ real-weight smoke test ≠ scientific benchmark.

Causality boundary — SAE learns sparse reconstruction features; crosscoder relates features across models; activation patching intervenes on internal values. Geometry correlations alone do not establish a causal mechanism. Separate personal implementation, experiments actually run, and team-level conclusions; historical speedups need their own logs/baselines.

Runbook — Pin inputs/environment → test adapter shapes → real-weight smoke check → assert paired IDs → tiny extract/reload → hand-check neighbors → self/shuffle controls → freeze selection roles → measure RAM/time → scale with restartable tasks. Seeds do not fix wrong row order, nondeterministic kernels or ties.

Source map — models/base.py, models/timeseries.py, experiments_layerwise.py; metrics/neighbors.py, kernel.py, physics.py; experiments/platonic/compute/pu_solve.py and slurm/pu_solve.sh; ARAVIND.md. Blog sections 1–19 retain derivations, figures, examples and 14 code walkthroughs.